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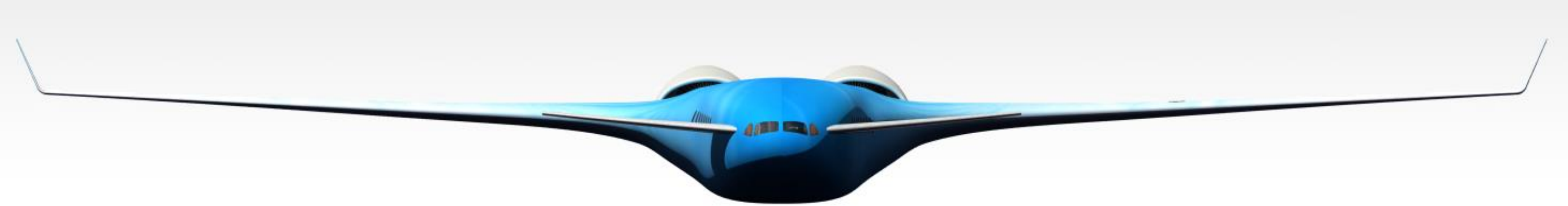

AE4238: Aero Engine Technology

Engine Inlets-II: Flowfield and Efficiency

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TU Delft

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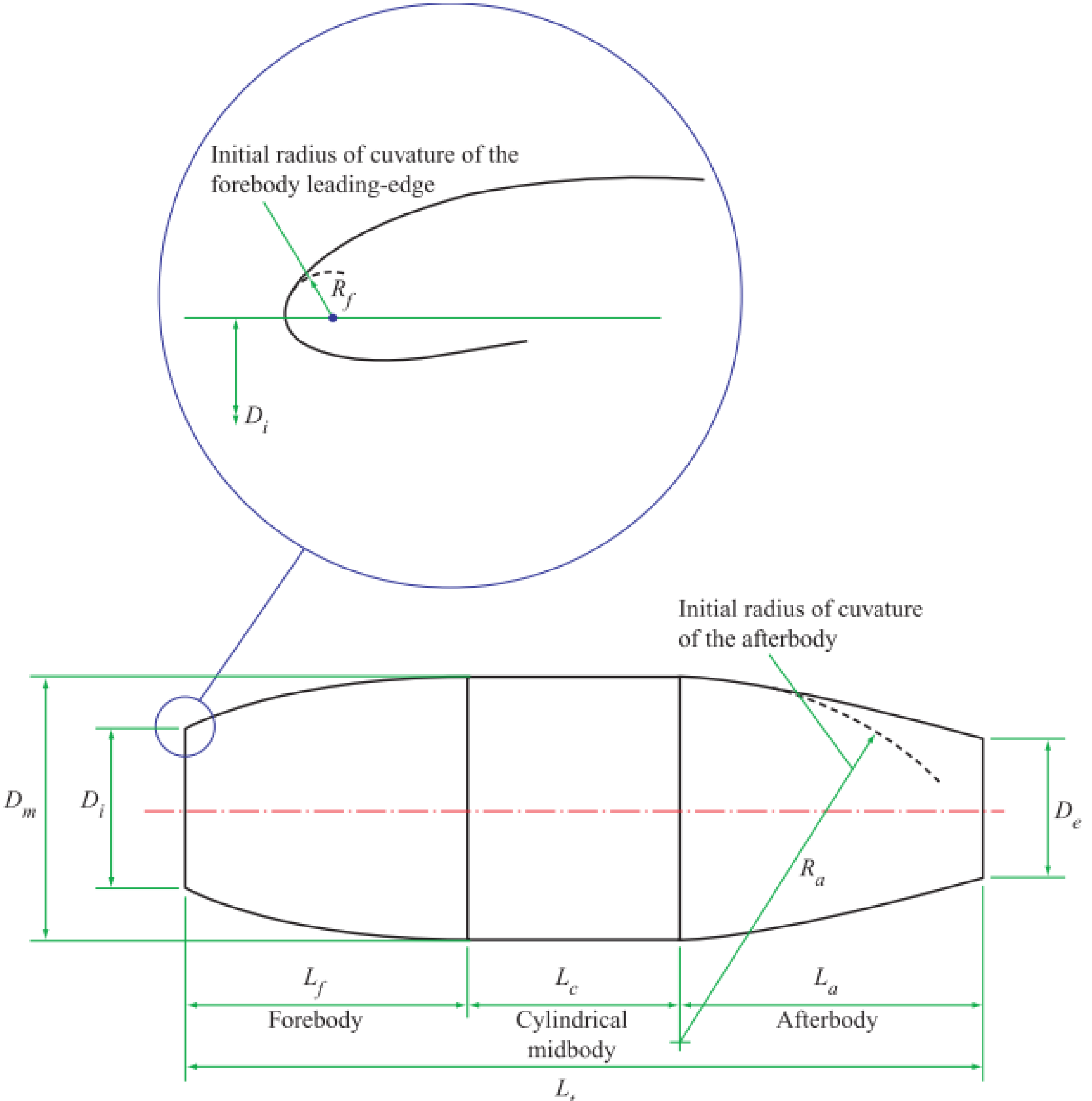
What will we learn in this lecture

- What are Aircraft Inlets
- Some of design criteria
- Examples

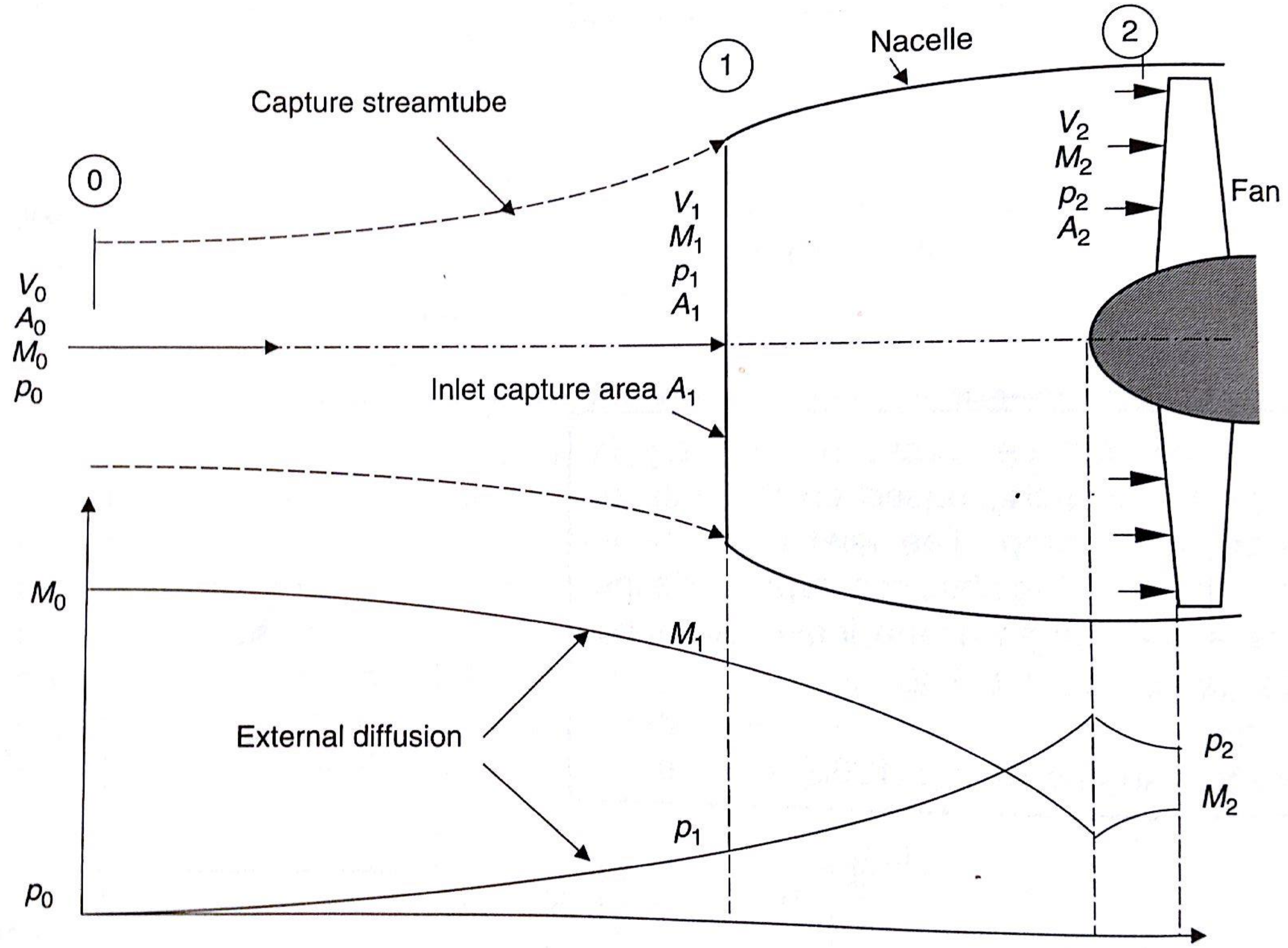
Engine Intake

Definition of cowling geometry



ESDU, *Drag of axisymmetric cowls at zero incidence for subsonic Mach numbers. Item no. 81024*. Engineering Sciences Data Unit, 1981

Variation of P and M in a typical inlet



■ **FIGURE 5.13** Typical subsonic inlet at cruise condition showing external as well as internal diffusion (note that flow accelerates over the fan centerbody)

Source: Aircraft Propulsion, Farokhi

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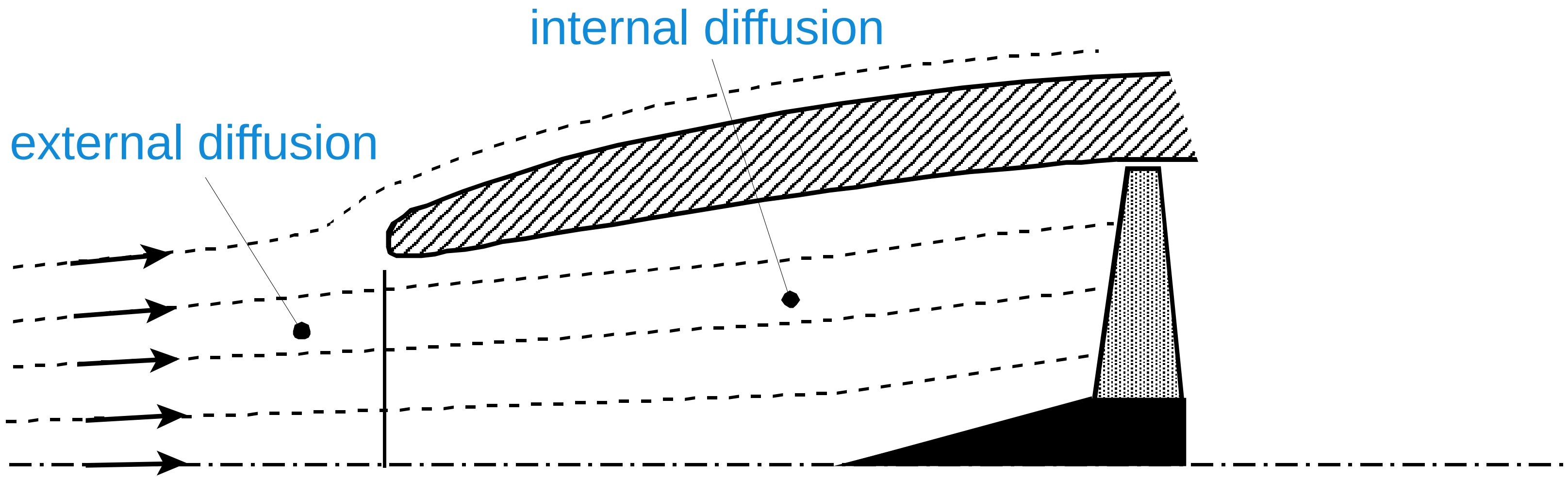
“Ram compression”

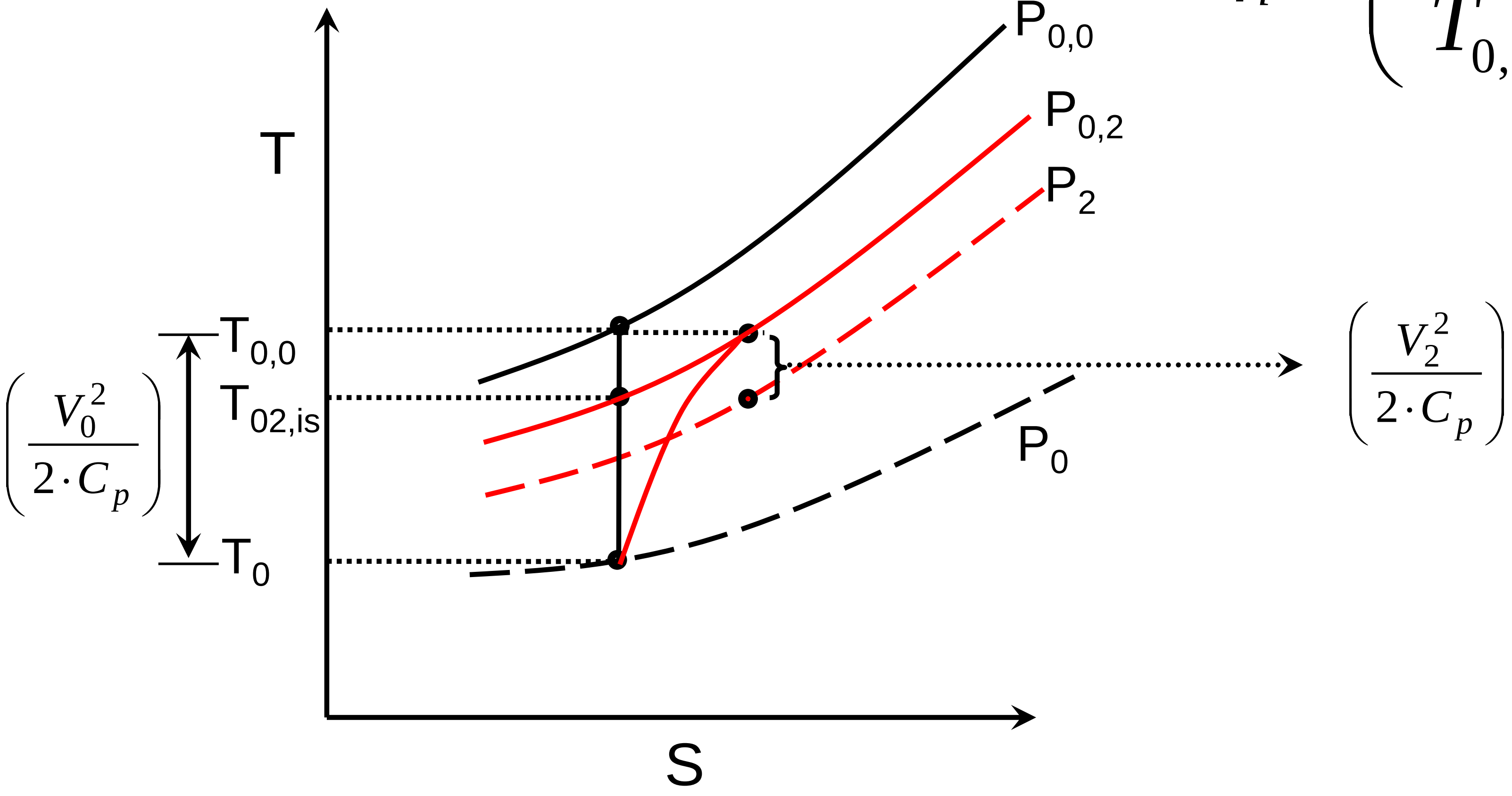
Ram compression

Aircraft engine inlet external and internal diffusion



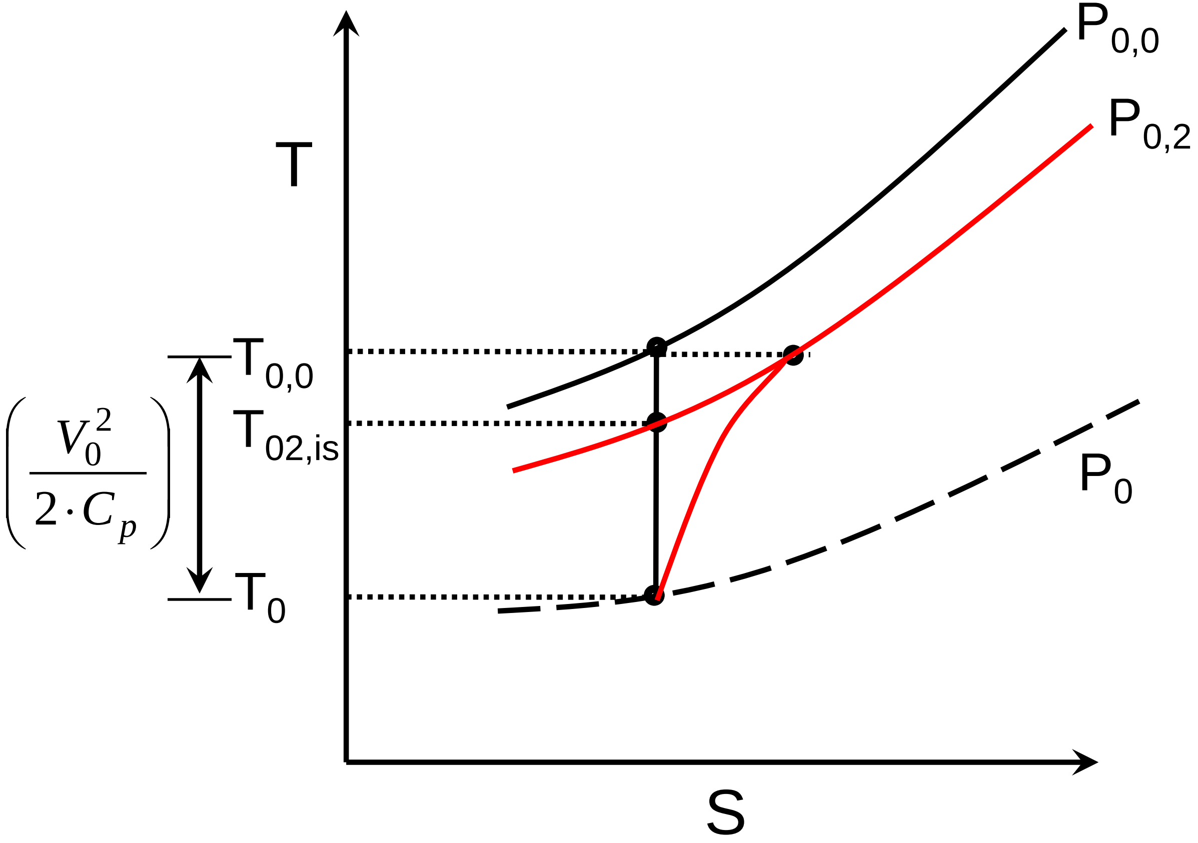
Inlet at Cruise

Flight speed > V_2



$$\eta_i = \left(\frac{T_{02,is} - T_0}{T_{0,0} - T_0}\right)$$

Ram Efficiency



Ram efficiency

$$\eta_{ram} = \frac{P_{0,2} - P_0}{P_{0,0} - P_0}$$

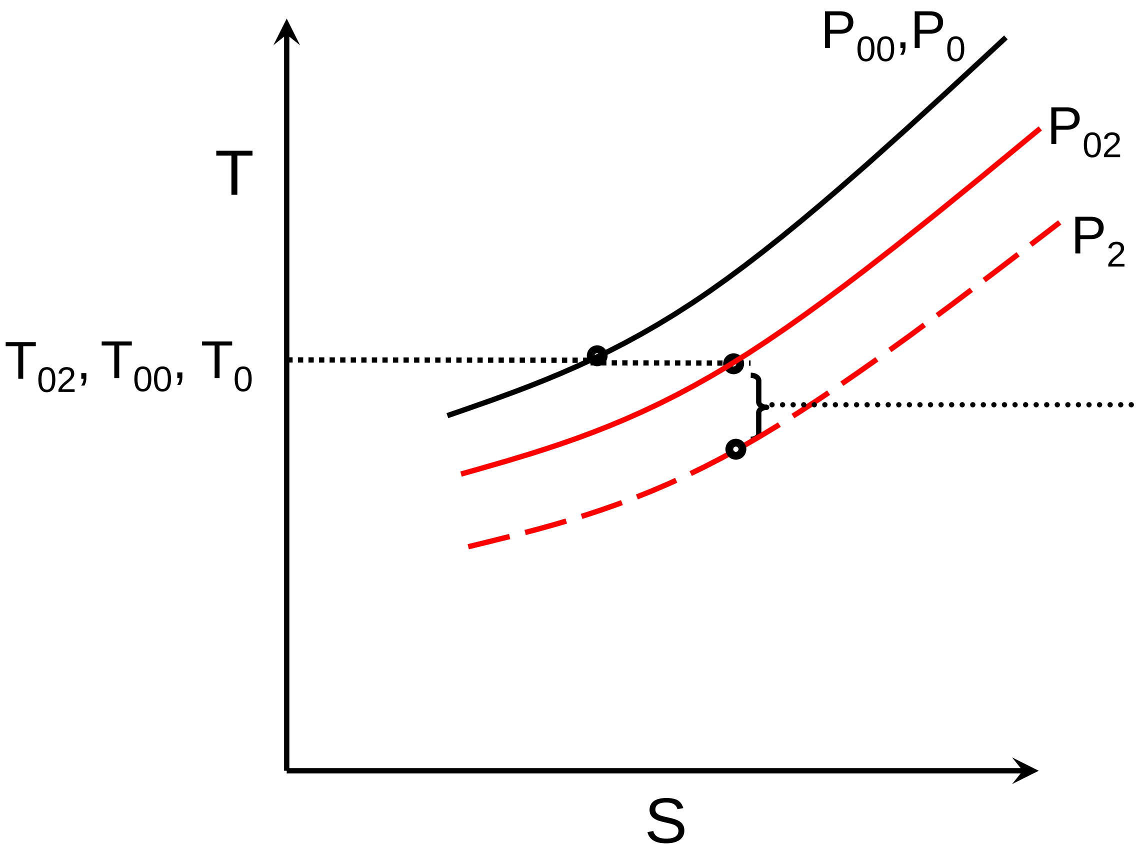
$$p_{0,2} = p_0 \left(1 + \eta_{is} \frac{v_0^{\ 2}}{2 c_p T_0}\right)^{\frac{k}{k-1}} = p_0 \left(1 + \eta_{is} \frac{k-1}{2} M_0^{\ 2}\right)^{\frac{k}{k-1}}$$

What happens at Take-Off ?

Inlet at Takeoff

$V_{flight} \cong 0$

$$PR = \left(\frac{P_{02}}{P_{00}}\right) = \left(\frac{P_{02}}{P_0}\right)$$



$$\left(\frac{V_2^2}{2 \cdot C_p}\right)$$

Engineering Practice

- η_{ram} is easier to measure
- η_{is} and η_{ram} can be used interchangeable till $M<0.8$
- For supersonic inlets "stagnation pressure ratio" or "pressure recovery factor" or "Ram Recovery Factor" is used = $P_{0,2}/P_{0,0}$
- RR (Ram Recovery) factor ~ 0.6 - 0.97
- Sometimes Empirical relations are used for supersonic inlets
  - $P_{02}/P_{00} = 1.0 - 0.075\ (M-1)^{1.35}$

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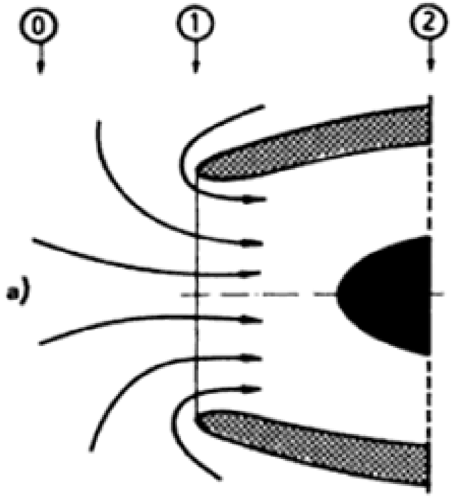
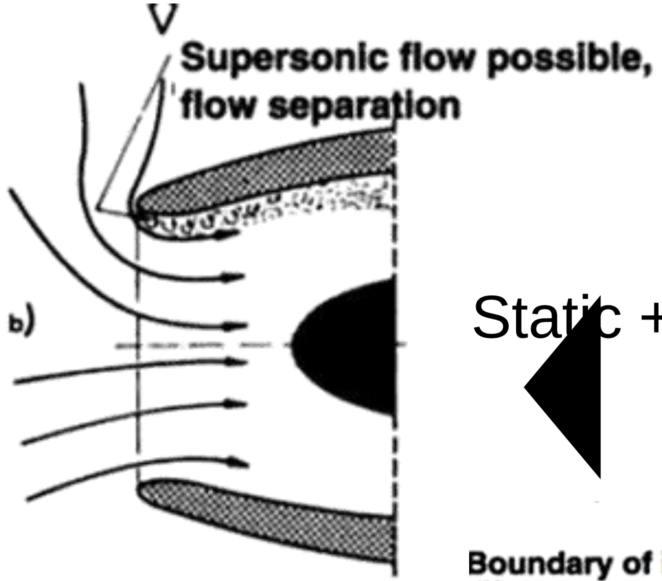
Subsonic intake @ offdesign

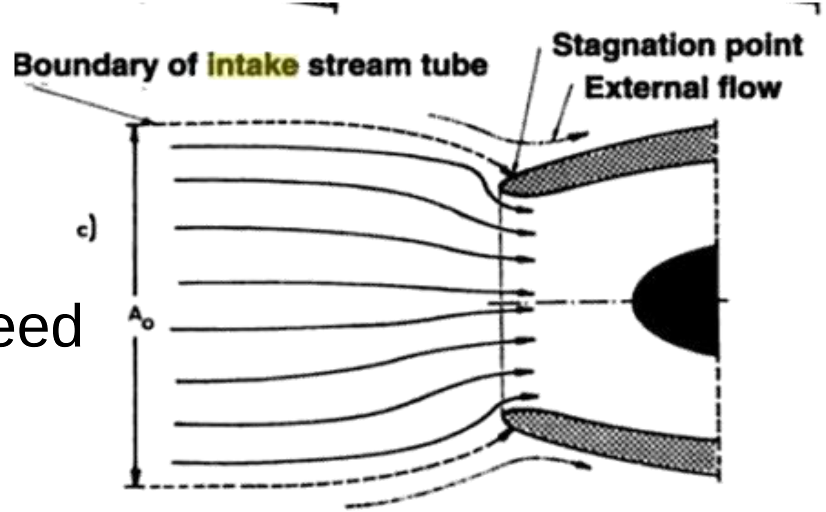


Aircraft at rest



Static + crosswind

Low Engine Speed



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Bell Mouth Intake

Picture: mdsaero

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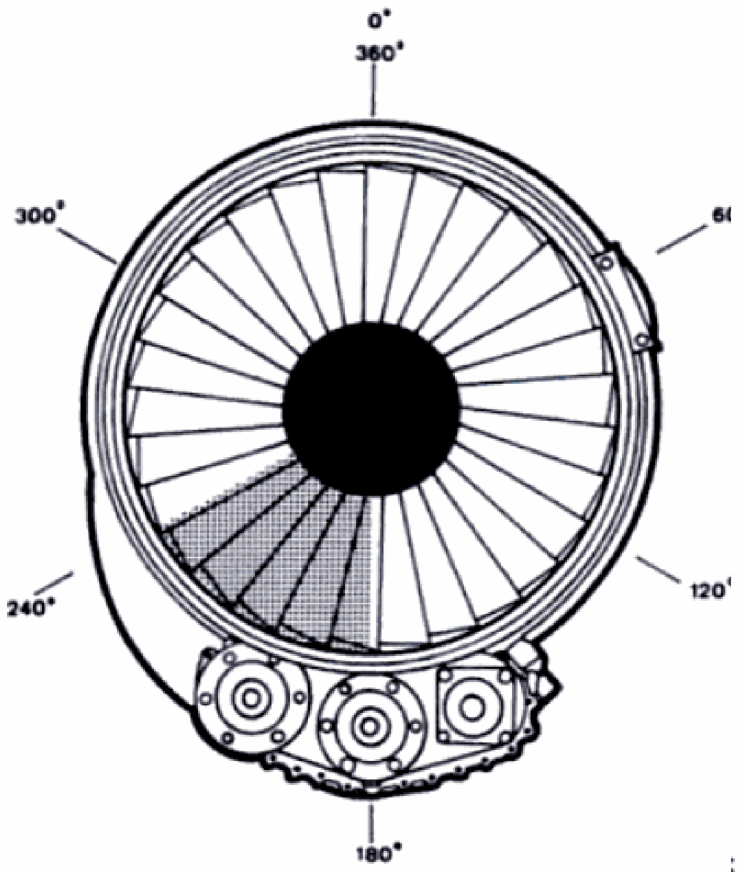
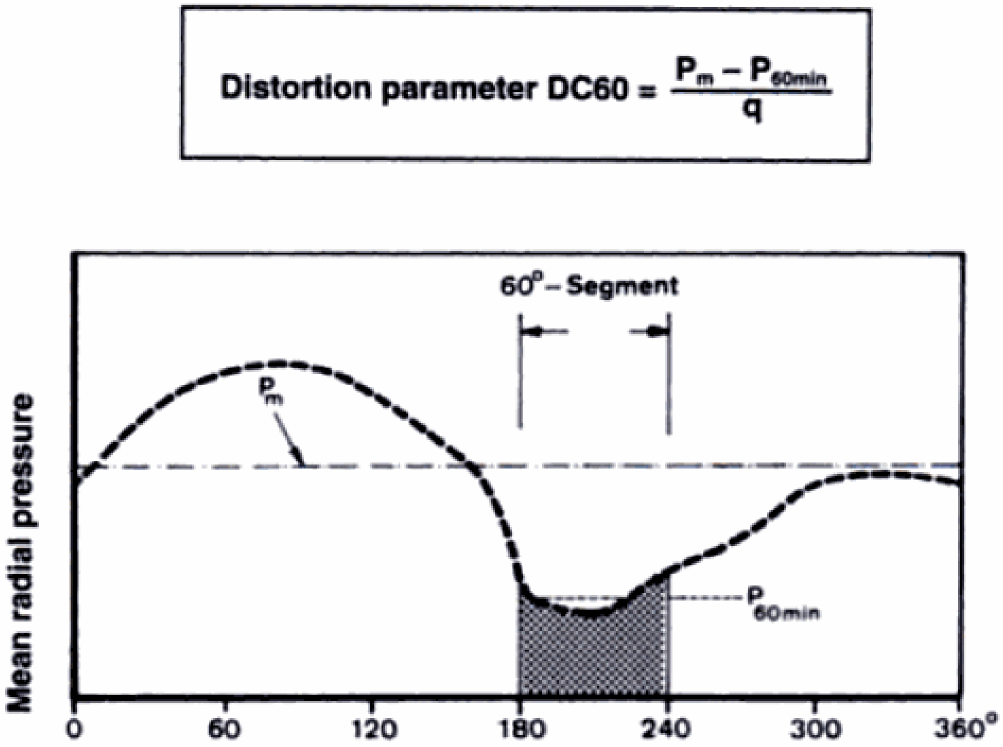
Distortion Parameter



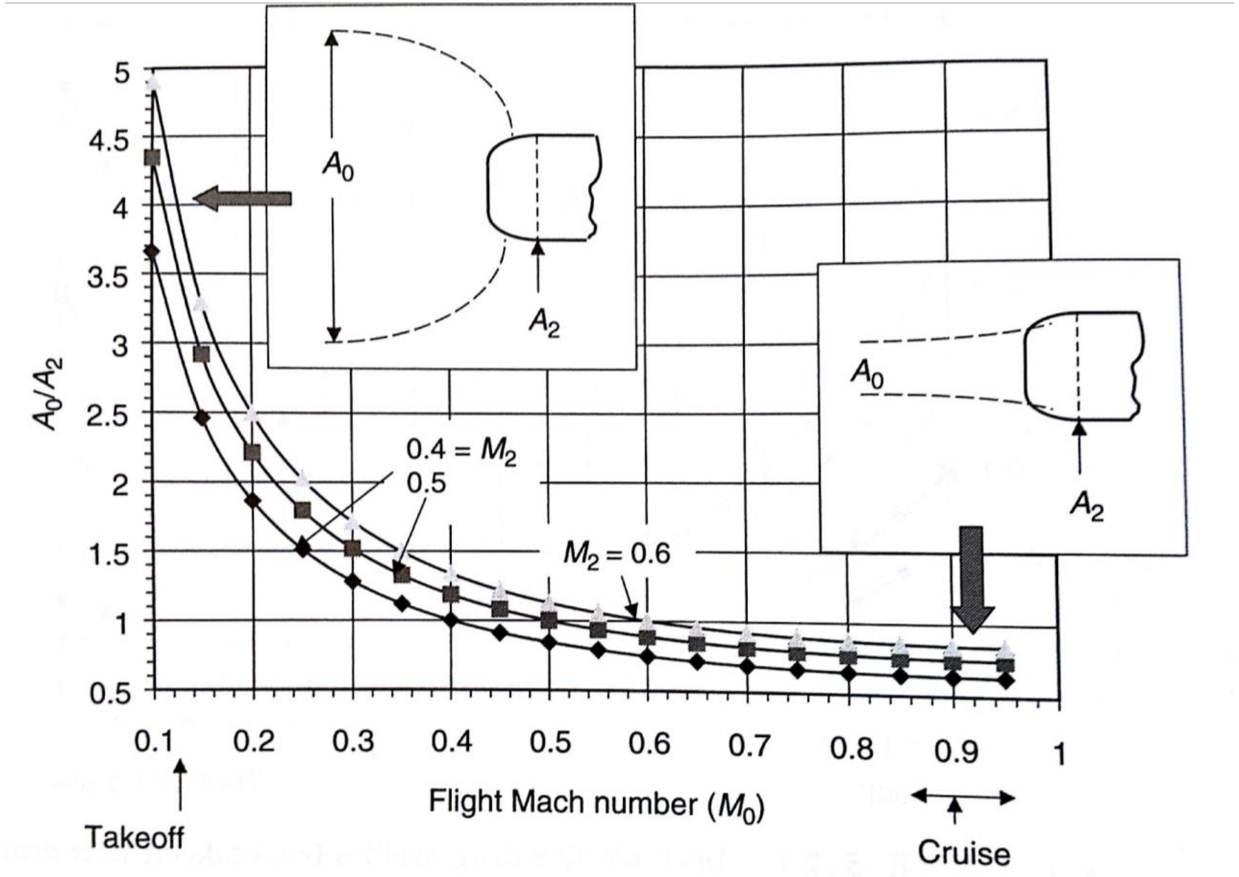


Explaining distortion factor

Surge (Video)

Distortion Parameter

Variation of capture area with Mach no.

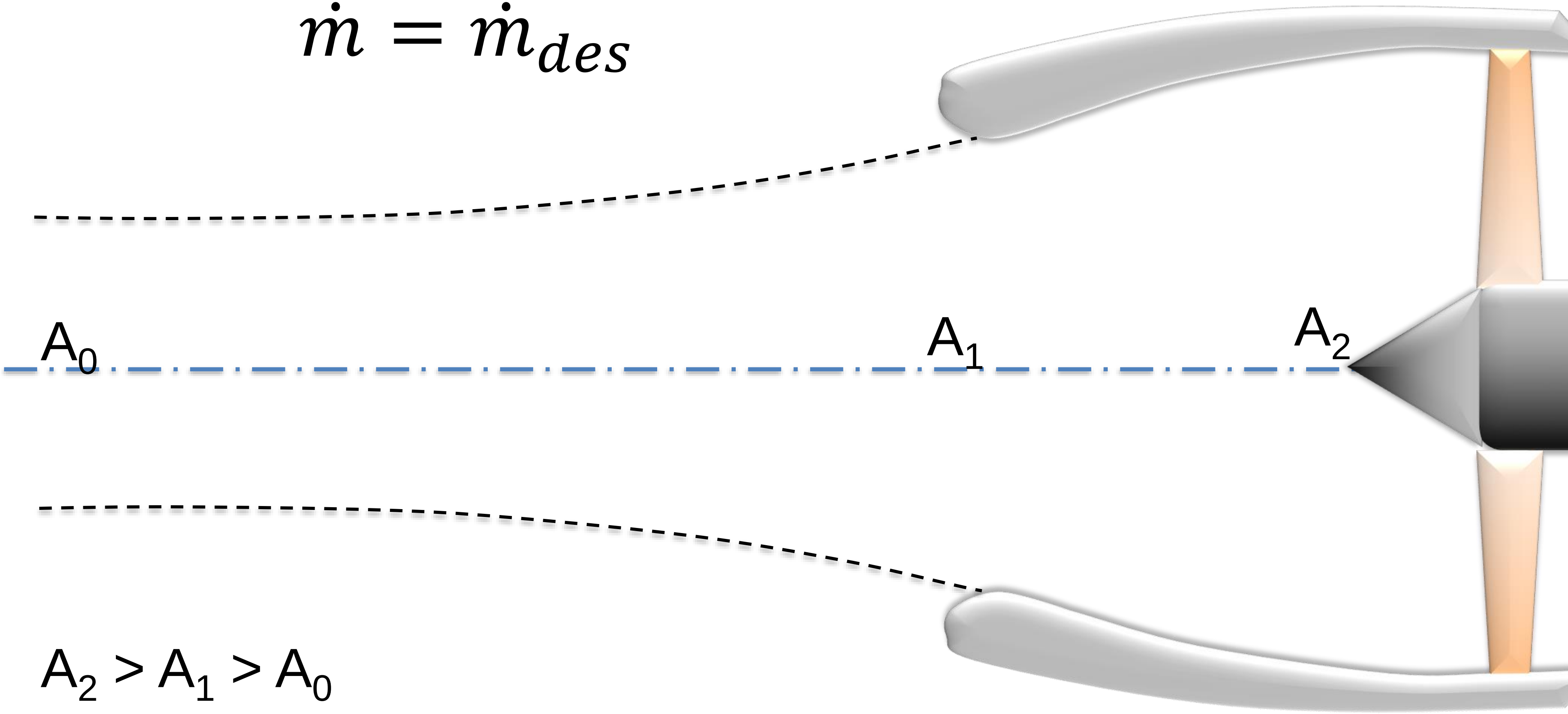


Captured stream-to-engine face area ratio from take-off to cruise

Source: Aircraft Propulsion, Farokhi

Flow field during cruise

$$\dot{m} = \dot{m}_{des}$$

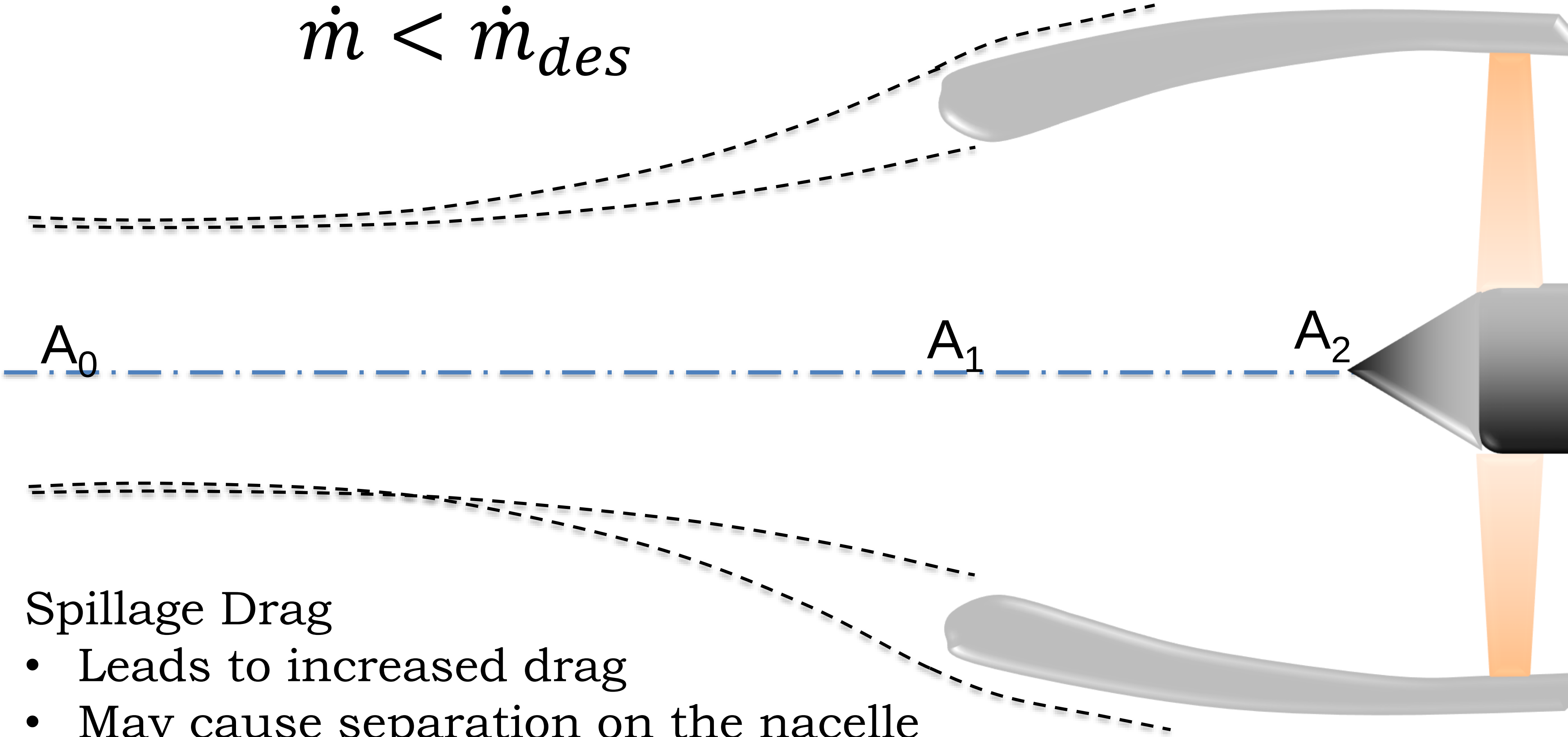


$A_2 > A_1 > A_0$

Note that part of the diffusion takes place outside the nacelle and is adiab

Flow field during cruise

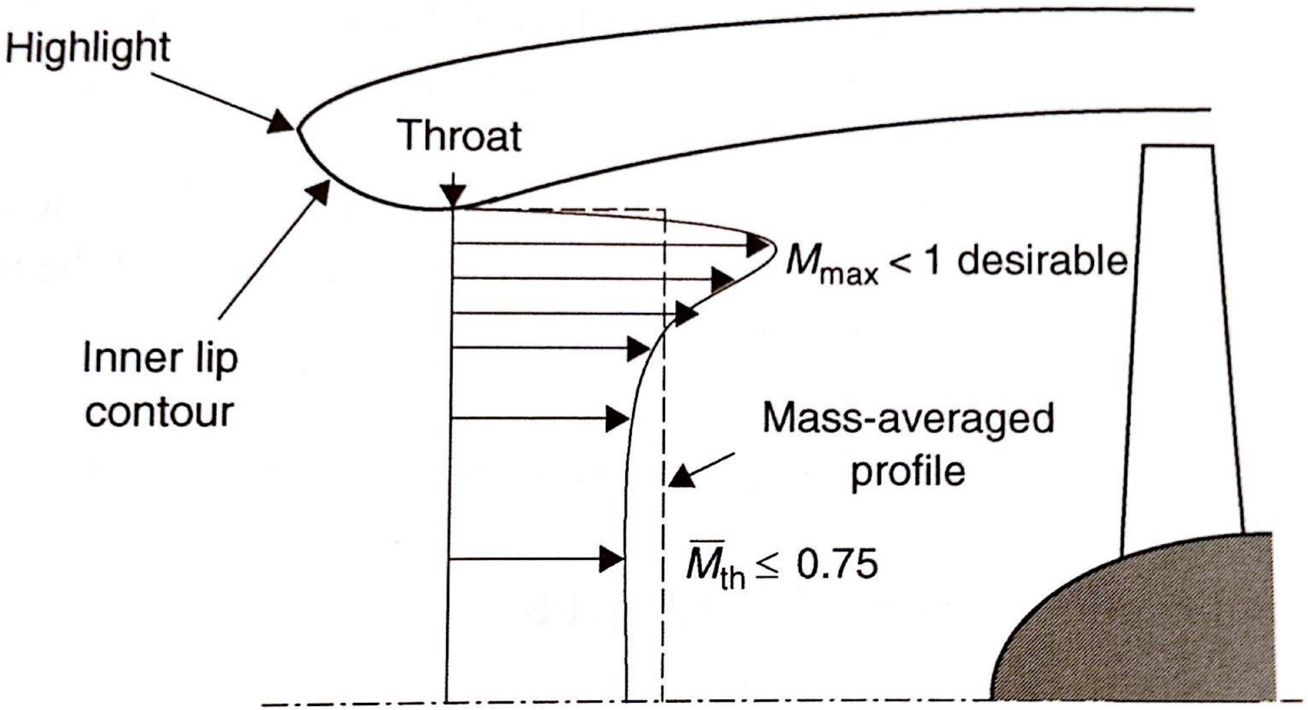
$$\dot{m} < \dot{m}_{des}$$



Spillage Drag

- Leads to increased drag
- May cause separation on the nacelle
- May lead to sonic bubble

Flow near the inlet lip



Flow curvature at the throat causes a non uniform velocity profile

Source: Aircraft Propulsion, Farokhi

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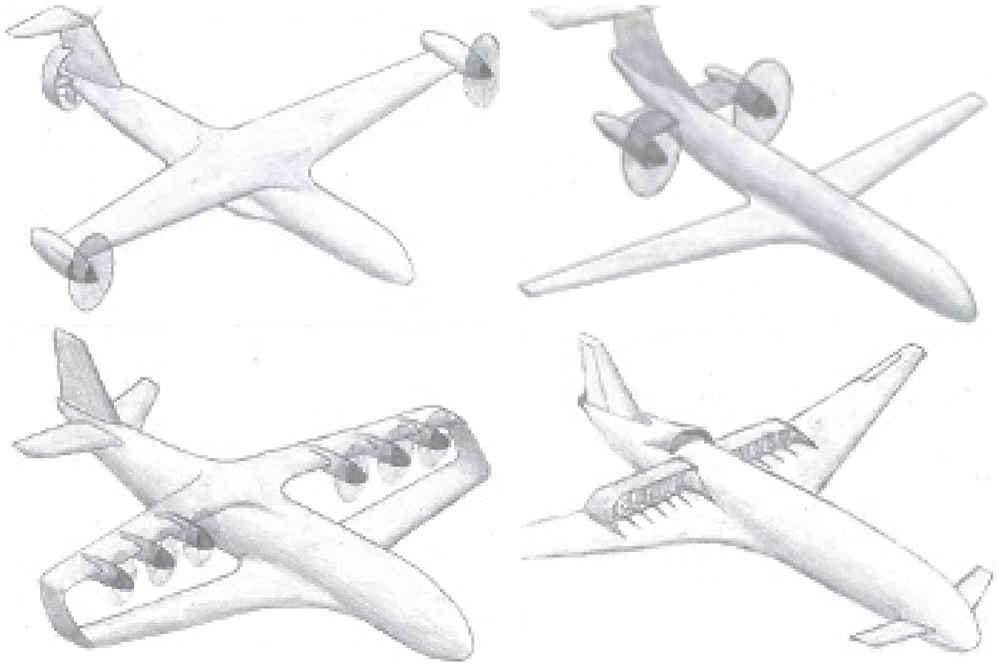
Thank You



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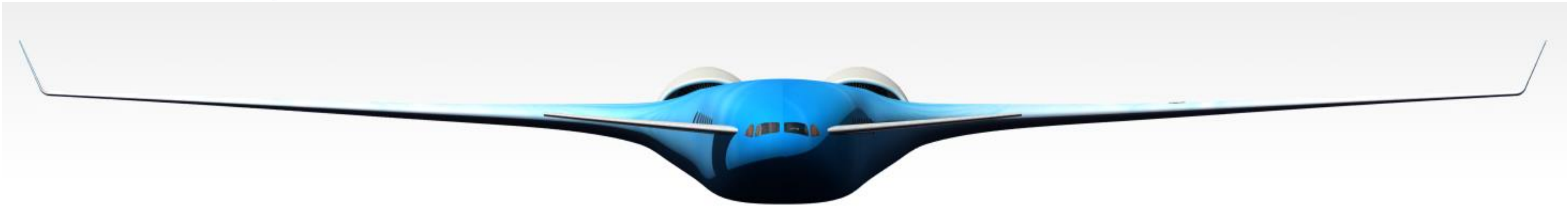

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